

Artificial Intelligent

LAB - 4

Fuzzy Logic and ANFIS

1. What is Fuzzy Logic?

Fuzzy Logic is an approach to computing based on degrees of truth rather than the usual true or false (1 or 0) logic used in Boolean systems.

Key Concept:

- In classical logic, a statement is either true (1) or false (0).
- In fuzzy logic, truth values range between 0 and 1.
 - Example:
 - Classical: “The water is hot” is either true or false.
 - Fuzzy: “The water is 70% hot” (truth value = 0.7).

Real-world Analogy:

- Human reasoning is fuzzy:
 - We say “it's somewhat cold,” “very fast,” or “mostly full” — these are fuzzy descriptions.

2. Why Fuzzy Logic?

- Handles uncertainty, imprecision, and approximate reasoning.
- Used where human-like reasoning is needed.
- Common in control systems, medical diagnosis, pattern recognition, etc.

3. Components of a Fuzzy Inference System (FIS)

Component	Description
Fuzzification	Converts crisp inputs to fuzzy values using membership functions
Rule Base	Set of IF-THEN rules (e.g., IF temperature is high THEN fan speed is high)
Inference Engine	Applies rules to fuzzy inputs to get fuzzy outputs
Defuzzification	Converts fuzzy outputs back to crisp values

4. Types of FIS

- Mamdani-type: Intuitive, used for control applications.
 - Sugeno-type: Uses linear or constant output functions. Suitable for optimization and learning.
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5. What is ANFIS? (Adaptive Neuro-Fuzzy Inference System)

ANFIS = Neural Network + Fuzzy Logic

It combines the learning capability of Neural Networks with the interpretability of Fuzzy Logic.

How ANFIS Works:

- Uses a Sugeno-type FIS.
- Learns fuzzy rules and membership function parameters from data.
- Uses hybrid learning algorithm (Least Squares + Gradient Descent).

Why Use ANFIS?

- Ideal when you have input-output data but no rule base.
 - Automatically generates and tunes fuzzy systems from training data.
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6. Differences: Fuzzy Logic vs ANFIS

Feature	Fuzzy Logic	ANFIS
Rule Definition	Manual	Learned from data
Learning Capability	No	Yes (via Neural Networks)
Output Type	Mamdani/Sugeno	Sugeno only
Use Case	Expert-defined systems	Data-driven modeling

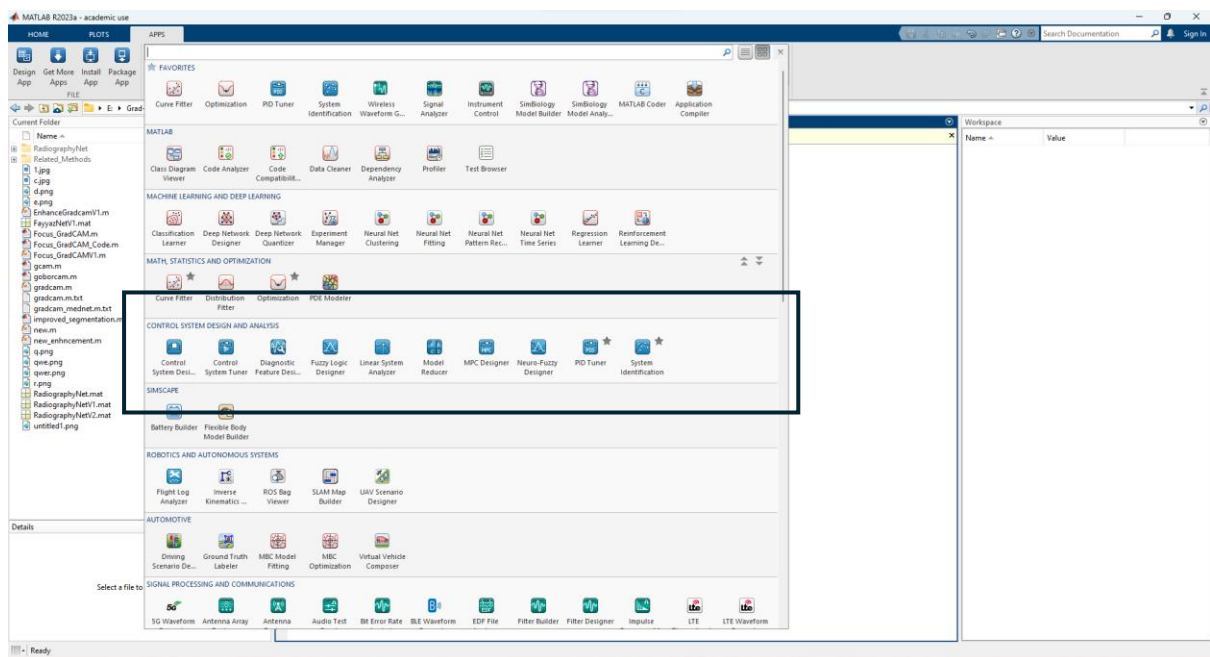
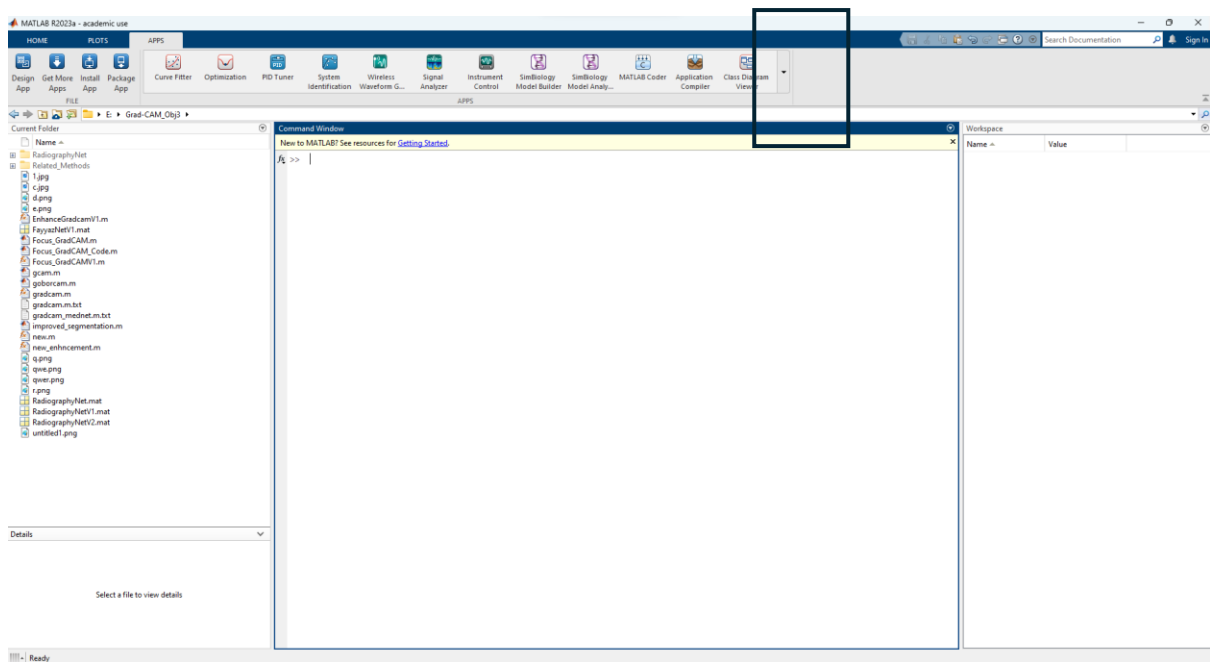
Step-by-Step Guide to Download MATLAB

Steps:

1. **Go to:** <https://www.mathworks.com>
2. **Click** “Sign In” at the top right.
3. **Create a MathWorks account** using your university email (e.g., yourname@utp.edu.my).
4. After verification, **log in** and go to:

🔗 <https://www.mathworks.com/downloads/>

5. Click **Download for Windows/Mac/Linux** (it detects your OS automatically).
6. **Install MATLAB Installer** and run it.
7. During installation:
 - Choose **“Log in with a MathWorks Account”**
 - Choose the license provided by your university
 - Select the **toolboxes** you need (e.g., Fuzzy Logic Toolbox, Deep Learning Toolbox)



Objectives:

- Understand fuzzy logic systems and their components.
- Learn how to implement fuzzy inference systems (FIS) in MATLAB.
- Train and simulate ANFIS models with real-world datasets.

Software Requirements:

- MATLAB (with Fuzzy Logic Toolbox)

Today's Lab

1. Introduction to Fuzzy Logic
2. Create a Basic FIS
3. Fuzzification and Membership Functions
4. Rule Definition and Evaluation
5. Surface View and Rule Viewer
6. FIS Using Sugeno-Type
7. Load and Modify FIS
8. ANFIS: Introduction and Use
9. Train ANFIS on Custom Dataset
10. ANFIS with Multiple Inputs
11. Predict New Values with Trained ANFIS
12. Export and Save FIS
13. ANFIS on Medical Dataset
14. ANFIS Error Analysis
15. Real-time Application Use Case

Code:

%% Example 1: Create a Basic FIS

```
fis = mamfis('Name', 'temperatureControl');  
disp(fis)
```

%% Example 2: Add Inputs, Outputs, and Membership Functions

```
fis = addInput(fis, [0 100], 'Name', 'Temperature');  
fis = addMF(fis, 'Temperature', 'trimf', [0 0 50], 'Name', 'Cold');  
fis = addMF(fis, 'Temperature', 'trimf', [25 50 75], 'Name', 'Warm');  
fis = addMF(fis, 'Temperature', 'trimf', [50 100 100], 'Name', 'Hot');
```

```

fis = addOutput(fis, [0 10], 'Name', 'FanSpeed');

fis = addMF(fis, 'FanSpeed', 'trimf', [0 0 5], 'Name', 'Low');

fis = addMF(fis, 'FanSpeed', 'trimf', [3 5 7], 'Name', 'Medium');

fis = addMF(fis, 'FanSpeed', 'trimf', [5 10 10], 'Name', 'High');

```

%% Example 3: Define Fuzzy Rules (Corrected Format)

```

ruleList = [

    1 1 1 1; % IF Cold THEN Low

    2 2 1 1; % IF Warm THEN Medium

    3 3 1 1 % IF Hot THEN High

];

fis = addRule(fis, ruleList);

```

%% Example 4: View Rules and Evaluate Output

```

% ruleviewer(fis);

output = evalfis(fis, 70);

disp(['Fan Speed at 70°C: ', num2str(output)]);

```

%% Example 5: Surface Viewer

```

gensurf(fis);

```

%% Example 6: Create a Sugeno-type FIS

```

fis2 = sugfis('Name','speedControl');

fis2 = addInput(fis2,[0 100],'Name','Speed');

fis2 = addOutput(fis2,[0 1],'Name','BrakeForce');

fis2 = addMF(fis2,'Speed','gaussmf',[10 20],'Name','Slow');

```

```

fis2 = addMF(fis2,'Speed','gaussmf',[10 80],'Name','Fast');
fis2 = addMF(fis2,'BrakeForce','linear',[0.01 0],'Name','Low');
fis2 = addMF(fis2,'BrakeForce','linear',[-0.01 1],'Name','High');
fis2 = addRule(fis2,[1 1 1 1; 2 2 1 1]);

```

```

%% Example 7: Load and Modify Existing FIS

```

```

fis3 = readfis('tipper');
showrule(fis3);

```

```

%% Example 8: ANFIS Training (Single Input)

```

```

data1 = [0 0; 5 2.5; 10 5; 15 7.5; 20 10];
anfisFIS1 = anfis(data1, 10);

```

```

%% Example 9: ANFIS with Two Inputs

```

```

data2 = [0 0 0; 5 2 2.5; 10 5 5; 15 7 7.5; 20 10 10];
anfisFIS2 = anfis(data2, 20);

```

```

%% Example 10: Predict with ANFIS

```

```

inputVal = [12 6];
predictedOutput = evalfis(anfisFIS2, inputVal);
disp(['Predicted Output: ', num2str(predictedOutput)]);

```

```

%% Example 11: Save and Reload Trained FIS

```

```

writefis(anfisFIS2, 'trainedFIS');
loadedFIS = readfis('trainedFIS');

```

```

%% Example 12: ANFIS on Medical Dataset (Diabetes) - Error-Free

```

```

% Load dataset (make sure diabetes.csv is in the current folder)

```

```
data3 = readmatrix('diabetes.csv'); % 8 features + 1 output (column 9)
```

```
% Normalize the data to [0, 1]
```

```
data3 = normalize(data3);
```

```
% Use only 2 features and 1 output to reduce rule complexity
```

```
dataSubset = data3(1:500, [1 2 9]); % Columns 1 & 2 as input, column 9 as output
```

```
% Train ANFIS model with 2 MFs per input
```

```
anfisFIS3 = anfis(dataSubset, 2);
```

```
%% Example 13: ANFIS Error Plot
```

```
[anfisFIS4, trainError] = anfis(dataSubset, 2); % Same data and MFs
```

```
figure;
```

```
plot(trainError);
```

```
title('ANFIS Training Error');
```

```
xlabel('Epochs');
```

```
ylabel('Error');
```

```
%% Example 14: Real-Time Application - Smart AC Control
```

```
% Training data: Temperature, Humidity → AC Intensity
```

```
data4 = [20 30 0.3;
```

```
         25 40 0.5;
```

```
         30 50 0.7;
```

```
         35 60 0.9;
```

```
         40 70 1];
```

```
% Train ANFIS with 2 MFs per input
```



```
anfisFIS5 = anfis(data4, 2);
```

```
% Evaluate new condition: 37°C and 65% humidity
```

```
result = evalfis(anfisFIS5, [37 65]);
```

```
disp(['AC Intensity for 37°C & 65% humidity: ', num2str(result)]);
```